Circular Motion

Answer key

Conventions used throughout

• **Uniform circular motion:** constant speed v around a circle of radius r. T = period (time for one trip around), f = frequency (trips per second), f = 1 ÷ T.

• **Centripetal force F_c:** not a new force. It is the net force toward the centre, made of the real forces (gravity, tension, normal force, friction). a_c = centripetal acceleration.

• **Forces:** F_g = gravity, F_N = normal force, T = tension, F_s = static friction.

• **Signs:** in the force questions, "toward the centre" is positive, so F_c = (forces toward the centre) − (forces away from it).

QUESTION 1

Is uniform circular motion accelerated?

1) Uniform circular motion is exactly what it sounds like – something moving in a circle at a constant speed.

a) Is an object in uniform circular motion accelerating? Explain.

b) Draw velocity vectors on the circle below to show the direction of the velocity at the points indicated.

c) The velocity is always _____ to the path and _____ to the radius of the circle.

a) Is an object in uniform circular motion accelerating? Explain.

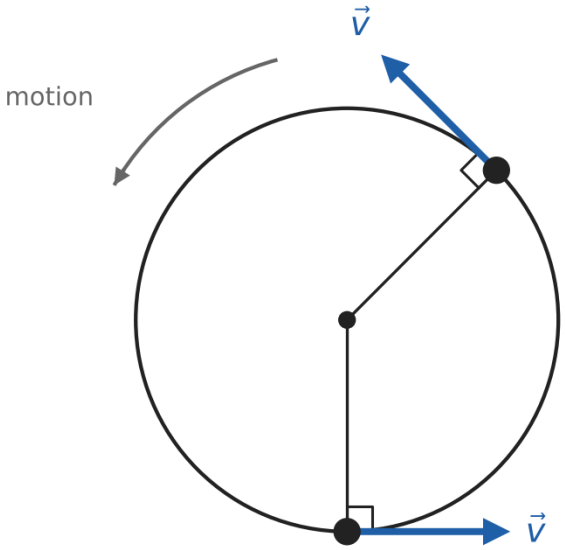
Yes. Velocity has a size and a direction. The speed stays the same, but the direction keeps changing as the object goes around. Any change in velocity is an acceleration:

$$\vec{a} = \frac{\Delta \vec{v}}{\Delta t} \neq 0 \quad \text{because the direction of } \vec{v} \text{ changes}$$

b) Draw velocity vectors on the circle below to show the direction of the velocity at the points indicated.

The arrow on the worksheet shows the object going counterclockwise. At each point the velocity points along the circle in that direction, at 90° to the radius.

Velocity vectors at the two points



c) The velocity is always _____ to the path and _____ to the radius of the circle.

Each velocity arrow just touches the circle (tangent) and makes a right angle with the radius (perpendicular).

FINAL ANSWER

a) Yes: the direction of the velocity keeps changing, so the velocity changes

b) Diagram above: each velocity is tangent to the circle, pointing the way the object moves

c) tangent (to the path); perpendicular (to the radius)

DEMONSTRATION

Discussion as a class

https://www.youtube.com/watch?v=ID0R43My4Co

Discussion as a Class:

1) When does he need to lift the hoop in order to knock the cards down?

2) In what direction is the net force? What does it say about the direction of acceleration?

1) When does he need to lift the hoop in order to knock the cards down?

While the hoop is down, it pushes the ball toward the centre and keeps it on the circle. The moment the hoop is lifted, that push stops and, by Newton's first law, the ball keeps going in a straight line in the direction it was moving: along the tangent. So he must lift the hoop when the ball's velocity (the tangent line) points straight at the cards, not when the ball is closest to them.

2) In what direction is the net force? What does it say about the direction of acceleration?

The hoop pushes the ball inward, so the net force points toward the centre of the circle. By Newton's second law the acceleration points the same way as the net force: toward the centre.

$$\Sigma\vec{F} = m\vec{a} \quad \Rightarrow \quad \vec{a} \text{ points the same way as } \Sigma\vec{F} \text{ (toward the centre)}$$

FINAL ANSWER

1) When the tangent to the circle at the ball points at the cards: the ball then travels straight along that tangent

2) The net force is toward the centre, so the acceleration is toward the centre too

FORCES 1

Ball on a string in a vertical loop

For each of the following draw a **Free Body Diagram** and state the force or forces that contribute to the centripetal force. Remember that for centripetal acceleration to occur, the **net force must be towards the centre of the circle**.

1) A ball is swinging on a string in a vertical loop. Consider the ball when

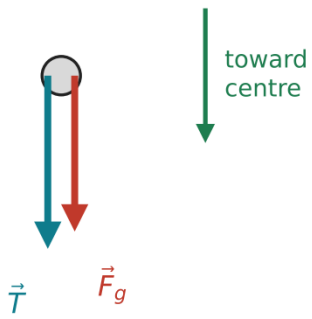
a. It is at the top of the loop

b. It is at the bottom of the loop

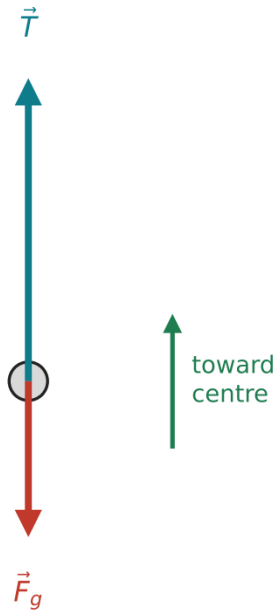
c. If the net centripetal force is to remain constant, what does that tell you about the tension in the string at the two different places?

FBDs of the ball (the string always pulls toward the centre)

a. top of the loop



b. bottom of the loop



a. It is at the top of the loop

At the top, the centre is below the ball. Tension (along the string, toward the centre) and gravity both point down, so both point toward the centre and both contribute:

$$F_c = T_{top} + F_g$$

b. It is at the bottom of the loop

At the bottom, the centre is above the ball. Tension points up (toward the centre) but gravity points down (away from the centre), so gravity works against tension:

$$F_c = T_{bottom} - F_g$$

c. If the net centripetal force is to remain constant, what does that tell you about the tension in the string at the two different places?

Set the two expressions for F_c equal and solve for T_{bottom}:

$$\text{Same } F_c: \quad T_{bottom} - F_g = T_{top} + F_g$$

$$\text{Add } F_g \text{ to both sides:} \quad T_{bottom} = T_{top} + 2F_g = T_{top} + 2mg$$

The string pulls harder at the bottom, by 2mg. At the bottom it must hold up the ball's weight and still supply F_c; at the top gravity helps, so the string pulls less.

FINAL ANSWER

a. Top: tension and gravity, both toward the centre ($F_c = T + F_g$)

b. Bottom: tension toward the centre minus gravity ($F_c = T - F_g$)

c. The tension is greater at the bottom: $T_{bottom} = T_{top} + 2mg$

Roller coaster loop

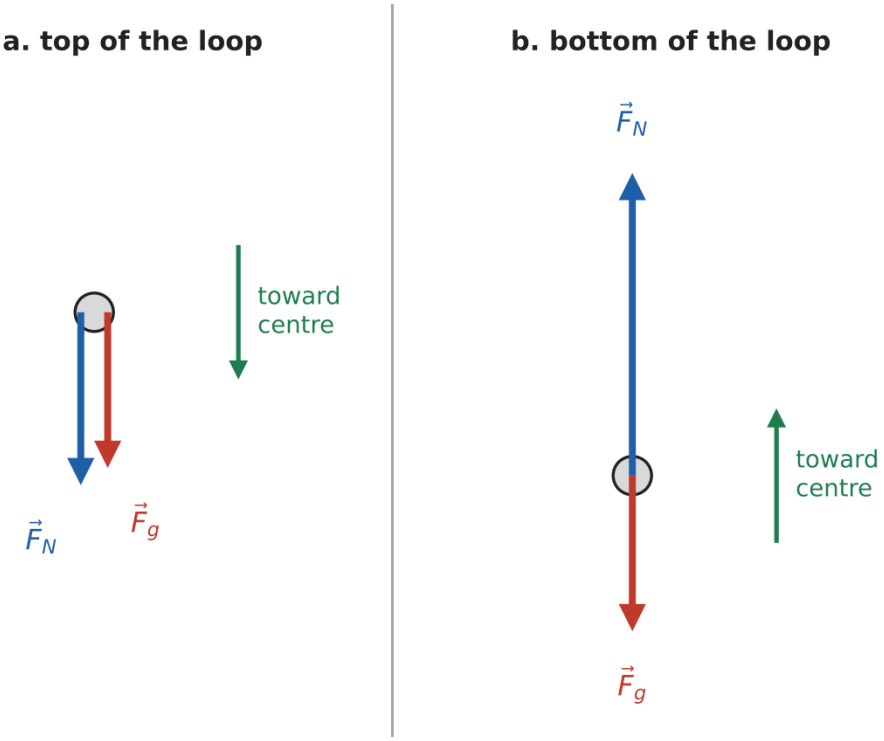
2) A roller coaster doing a loop the loop. Consider the train when

a. It is at the top of the loop

b. It is at the bottom of the loop

c. At which location would a person feel heavier? Explain.

FBDs of the train (the track pushes it toward the centre)



a. It is at the top of the loop

The train is upside down on the inside of the track, so the track pushes it down, toward the centre. Gravity also points down:

$$F_c = F_N + F_g$$

b. It is at the bottom of the loop

The track pushes up (toward the centre) and gravity pulls down (away from the centre):

$$F_c = F_N - F_g$$

c. At which location would a person feel heavier? Explain.

How heavy you feel is the normal force (2.6). Solve each equation for F_N:

$$\text{Top:} \quad F_N = F_c - mg \qquad \text{Bottom:} \quad F_N = F_c + mg$$

At the bottom the seat has to hold you up against gravity **and** push you toward the centre, so F_N is biggest there and you feel heavier. At the top gravity does part of the job, so you feel lighter.

FINAL ANSWER

a. Top: normal force and gravity, both toward the centre ($F_c = F_N + F_g$)

b. Bottom: normal force minus gravity ($F_c = F_N - F_g$)

c. At the bottom, where the normal force is largest

FORCES 3

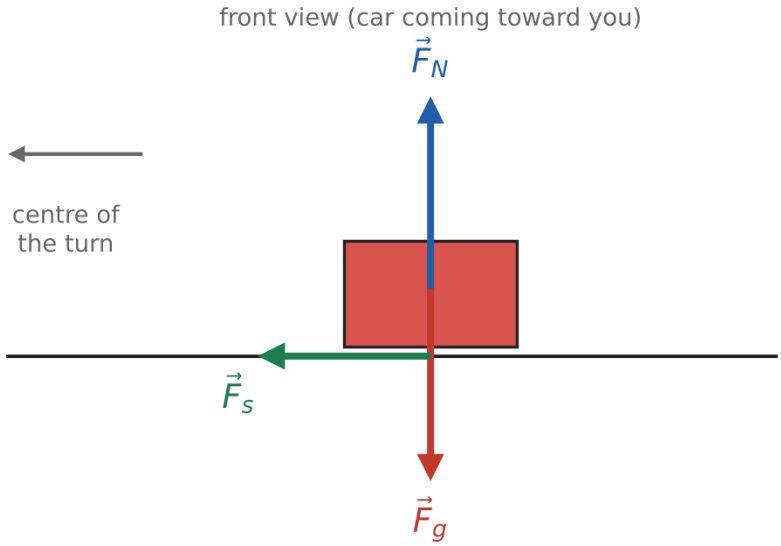
Car turning on a level road

3) A car is turning on a level road. (coming toward you in the picture)

Gravity (down) and the normal force (up) cancel, because the car doesn't move up or down. The only sideways force is static friction from the road on the tires, pointing toward the centre of the turn. (The tires grip and don't slide sideways, so the friction is static.)

$$F_c = F_s$$

FBD of the car (front view)



FINAL ANSWER

Static friction from the road provides the centripetal force: $F_c = F_s$

FORCES 4

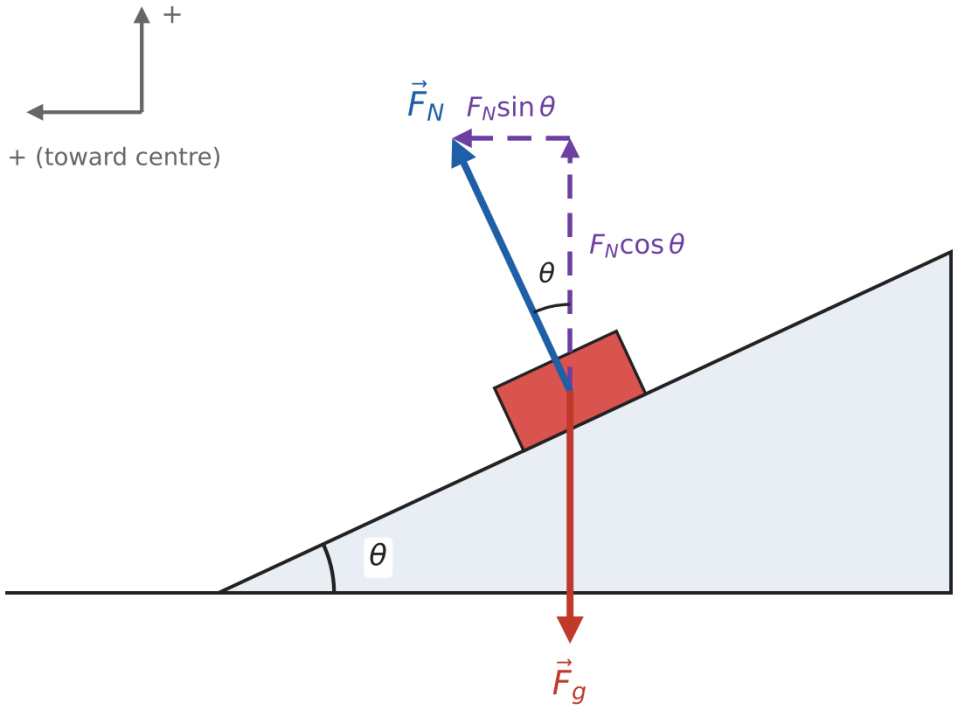
Car turning on a banked road

4) A car is turning on a **banked** road. (coming toward you in the picture). Draw the components of the appropriate force(s) according to the co-ordinate system given in the picture. Would it be possible to corner safely without static friction? Explain.

The worksheet's axes are vertical (+ up) and horizontal (+ toward the centre of the turn). Gravity is already vertical. The normal force is perpendicular to the banked road, so it tilts toward the centre by the bank angle θ. Split it into components (dashed):

Toward the centre: $F_{Nx} = F_N \sin\theta$ Up: $F_{Ny} = F_N \cos\theta$

FBD of the car on the banked road (no friction)



Would it be possible to corner safely without static friction?

Yes, but only at one speed. Without friction, the only force with a part toward the centre is the normal force, so F_N sin θ must supply all of F_c. Vertically, F_N cos θ holds up the car:

Up/down:

$$F_N\cos\theta - mg = 0 \qquad \text{Toward the centre:} \quad F_N\sin\theta = \frac{mv^2}{r}$$

For a given bank angle and radius, these work together for exactly one speed (the speed the curve is designed for). Faster than that, the car needs friction pointing down the bank to stop it sliding up and out; slower, it needs friction pointing up the bank to stop it sliding down and in.

FINAL ANSWER

F_N sin θ (toward the centre) and F_N cos θ (up); F_g is straight down

Yes, but only at one particular speed, where F_N sin θ alone gives exactly the needed F_c. At any other speed static friction is needed.

DERIVATION

Calculating centripetal acceleration

Goal: To derive an equation that will allow us to calculate the magnitude of the instantaneous centripetal acceleration in terms of easily measurable quantities: speed (*v*) and radius of curvature (*r*)

Basic Idea: We will start with the regular equation for acceleration, $\vec{a} = \Delta\vec{v}/\Delta t$, and just consider a very small time interval.

a) Consider an object moving from position $\vec{r}_1$ to position $\vec{r}_2$ as shown in the diagram. Start with the basic definition of acceleration above. We know that when vectors are not co-linear, we have to draw a vector diagram to determine $\Delta\vec{v} = \vec{v}_2 - \vec{v}_1$. Do this below. Label it:

b) Now we need to get a connection between *v* and *r*, so draw another vector diagram to show $\Delta\vec{r} = \vec{r}_2 - \vec{r}_1$. Label it clearly.

A couple of simplifications: the magnitude of velocity never changes and neither does the magnitude of the radius. Hence, we can drop our subscripts *i* and *f* ($v_i = v_f = v$ and $r_i = r_f = r$)

c) If the triangles are similar then finish the ratio: $\Delta v/v =$ _____

d) Now isolate Δv in the ratio above:

e) Take your expression from (d) and sub it into the basic acceleration equation:

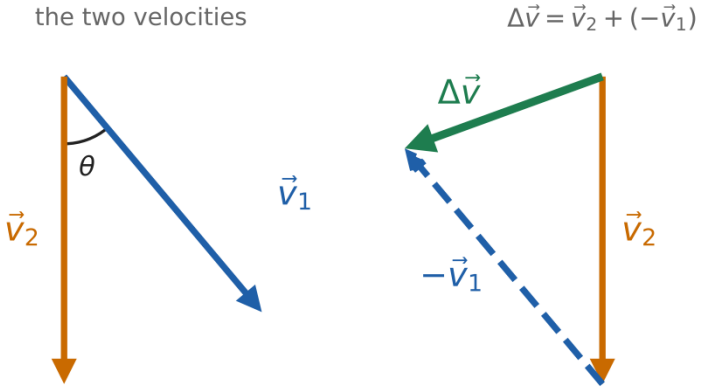
f) Notice that $\vec{r}$ is the position vector for the object, so $\Delta\vec{r}$ is change in position which makes $\Delta\vec{r}/\Delta t$ equal to speed which is *v*. Make this substitution in your current acceleration equation and simplify. Check your final result with the textbook. (p 116)

a) Draw a vector diagram to determine $\Delta\vec{v} = \vec{v}_2 - \vec{v}_1$.

In the worksheet's diagram the object moves clockwise, so $\vec{v}_1$ points down and to the right and $\vec{v}_2$ points straight down. Subtracting a vector means adding its opposite: draw $\vec{v}_2$, then $-\vec{v}_1$ tip to tail. $\Delta\vec{v}$ goes from the first tail to the last tip. Each velocity is perpendicular to its radius, so the angle between $\vec{v}_1$ and $\vec{v}_2$ is the same θ as between the two radii.

$\Delta\vec{v}$ points toward the centre of the circle: that is the direction of the acceleration.

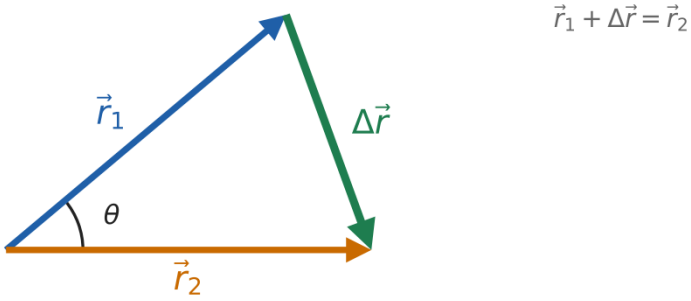
Vector diagram for $\Delta\vec{v}$



b) Draw another vector diagram to show $\Delta\vec{r} = \vec{r}_2 - \vec{r}_1$.

$\Delta\vec{r}$ goes from the tip of $\vec{r}_1$ to the tip of $\vec{r}_2$, because $\vec{r}_1 + \Delta\vec{r} = \vec{r}_2$.

Vector diagram for $\Delta\vec{r}$



c) If the triangles are similar then finish the ratio: Δv/v = _____

Both triangles have two equal sides with the same angle θ between them (v and v; r and r), so they are similar. Matching sides are in the same ratio:

$$\frac{\Delta v}{v} = \frac{\Delta r}{r}$$

d) Now isolate Δv in the ratio above:

Multiply both sides by v: $\Delta v = \frac{v\,\Delta r}{r}$

e) Take your expression from (d) and sub it into the basic acceleration equation:

Formula: $a = \frac{\Delta v}{\Delta t}$

Put in $\Delta v = \frac{v\,\Delta r}{r}$: $a = \frac{v\,\Delta r}{r\,\Delta t}$

f) Make this substitution in your current acceleration equation and simplify.

Write the fraction as v ÷ r times Δr ÷ Δt, then put in Δr ÷ Δt = v:

$$a = \frac{v}{r} \cdot \frac{\Delta r}{\Delta t}$$

$$\text{Put in } \frac{\Delta r}{\Delta t} = v: \quad a_c = \frac{v}{r} \cdot v = \frac{v^2}{r}$$

FINAL ANSWER

a) $\Delta\vec{v} = \vec{v}_2 + (-\vec{v}_1)$, pointing toward the centre (diagram above)

b) $\Delta\vec{r}$ joins the tip of $\vec{r}_1$ to the tip of $\vec{r}_2$ (diagram above)

c) $\Delta v/v = \Delta r/r$

d) $\Delta v = v\Delta r/r$

e) $a = v\Delta r/(r\Delta t)$

f) $a_c = v^2/r$

ALTERNATIVE FORMULAS

Using the period or the frequency

Sometimes you know the frequency or period of an object instead of its speed. We therefore need equations for centripetal acceleration that include frequency or period. We will derive these:

Consider an object moves at a constant speed v around a circle of radius r in a time T (called the period since circular motion is periodic).

a) You know **v = Δd/Δt**. For an object moving in a circle of radius r, what is the distance travelled? What is the time taken for one revolution? Write the formula for the speed of the object in terms of r and T.

b) Sub your formula into the centripetal acceleration formula and simplify. Compare your result with your textbook, page 116 bottom.

c) Frequency is the reciprocal of period. Modify your formula in b) to use frequency instead of period. Check your results.

a) Write the formula for the speed of the object in terms of r and T.

In one revolution the object goes once around the circumference, 2πr, and that takes one period, T:

Formula: $v = \frac{\Delta d}{\Delta t}$

Put in $\Delta d = 2\pi r,\ \Delta t = T$: $v = \frac{2\pi r}{T}$

b) Sub your formula into the centripetal acceleration formula and simplify.

Formula: $a_c = \frac{v^2}{r}$

Put in $v = \frac{2\pi r}{T}$: $a_c = \frac{\left(\frac{2\pi r}{T}\right)^2}{r}$

Square the top: $a_c = \frac{4\pi^2 r^2}{T^2 r}$

Cancel one r: $a_c = \frac{4\pi^2 r}{T^2}$

c) Modify your formula in b) to use frequency instead of period.

Frequency is the reciprocal of period, so T = 1 ÷ f:

Put in $T = \frac{1}{f}$: $a_c = \frac{4\pi^2 r}{\left(\frac{1}{f}\right)^2} = \frac{4\pi^2 r}{\frac{1}{f^2}}$

Dividing by $\frac{1}{f^2}$ is multiplying by f^2: $a_c = 4\pi^2 r f^2$

FINAL ANSWER

a) Distance 2πr, time T, so v = 2πr/T

b) $a_c = 4\pi^2 r/T^2$

c) $a_c = 4\pi^2 r f^2$